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MDMA-assisted psychotherapy for AUD: Bayesian analysis of WHO drinking risk level and exploratory analysis of drinking behavior and psychosocial functioning at 3 months follow-up

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Abstract

Aims: Safety and tolerability data from the first open-label feasibility study of 3,4-methylenedioxymethamphetamine (MDMA)-assisted psychotherapy for alcohol use disorder was recently published. This paper presents a Bayesian analysis of the impact of MDMA-assisted psychotherapy on treatment success, defined as 2-level reduction in the World Health Organization (WHO) drinking risk at the 3 months follow-up. We also examined the impact on drinking behavior and psychosocial measures at 3 months compared to baseline.
Methods: Fourteen participants with a diagnosis of alcohol use disorder who had recently undergone detoxification completed an eight-week course of ten psychotherapy sessions, including two sessions with MDMA. Measures assessing drinking behavior, quality of life, sleep, self-compassion, and empathy were collected. Bayesian analysis using flat and skeptical priors was performed to determine treatment success defined as a 2-level reduction in WHO drinking risk.
Results: Bayesian analysis suggested that the probability of a 2-level reduction in WHO drinking risk from baseline to 3 months post-treatment is 55%–63%, based upon either a flat or skeptical prior respectively. We present preliminary findings suggesting reductions in alcohol craving (measured by the Penn Alcohol Craving Scale and Obsessive Compulsive Drinking Scale) and improvements in sleep and aspects of psychosocial functioning at 3 months follow-up compared to baseline.
Conclusions: The Bayesian analysis provides a useful harm reduction endpoint interpretation of drinking in terms of a 2-level reduction in WHO drinking risk. Further findings provide preliminary insights into the potential impact of MDMA-assisted psychotherapy on quality of life and well-being in addition to reductions in drinking.
ClinicalTrials.gov ID: NCT04158778.

Keywords: MDMA; psychotherapy; well-being; craving; psychosocial; sleep

Introduction

The use of 3,4-methylenedioxymethamphetamine (MDMA)-assisted psychotherapy was initially explored in the 1980s (Greer and Tolbert 1998) and has been followed by the recent resurgence of clinical research led by the Multidisciplinary Association for Psychedelic Studies (MAPS) (Mitchell et al. 2021, 2023). Currently results from the multi-site, randomized, placebo-controlled, Phase 3 trials demonstrate substantial efficacy of MDMA-assisted therapy for moderate to severe post-traumatic stress disorder (PTSD) (Mitchell et al. 2021, 2023). These results suggest that MDMA-assisted psychotherapy may be a promising treatment strategy in other psychiatric disorders associated with trauma, including alcohol use disorder (AUD) (Brady and Back 2012, McCauley et al. 2012, Smith and Cottler 2018). PTSD is commonly associated with the co-occurrence of AUD and substance use disorders (McCauley et al. 2012). Interestingly further analysis from the MAPS Phase 3 trial suggests MDMA-assisted psychotherapy for severe PTSD may also lead to subclinical improvements in alcohol use (Nicholas et al. 2022).

The first open-label trial of MDMA-assisted psychotherapy for AUD provided preliminary evidence supporting its safety and tolerability as a treatment approach (Sessa et al. 2021). The study enrolled 14 participants in the eight-week psychotherapy course consisting of two dosing sessions with MDMA-assisted psychotherapy. MDMA was well-tolerated and no unexpected adverse events were observed. The study reported reductions in drinking: at the 9 months follow-up 11 participants were drinking <14-week psychotherapy course consisting 9 who were totally abstinent from alcohol), while 3

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participants had relapsed to drinking >14 cited adverse events were observed. The study reported improvements in anxiety and depression at 3, 6 and 9 months follow-up compared to baseline and screening. There is interest in the extent to which MDMA-assisted psychotherapy may impact drinking behavior and improve general measures of health and well-being that are not condition-specific.

Alcohol use remained low at 9 months following detoxification in the original Sessa et al. 2021 study. Reductions in WHO drinking risk levels for alcohol consumption, rather than complete abstinence, appears to be achievable, is associated with better functioning, and is maintained over time in both the United States and the United Kingdom (Witkiewitz et al. 2020). These reductions offer a harm reduction endpoint and framework whereby a 2-level decrease acts as a meaningful benchmark associated with reductions in alcohol-related morbidity and healthcare costs in United States and the United Kingdom (an aspect not previously reported in the Sessa et al. 2021 study). Furthermore, an important factor in mediating relapse that was also not reported in the Sessa et al. 2021 study is craving. Cravings in AUD refers to the strong and intense desires or urges to consume alcohol, driven by psychological and physiological mechanisms. Higher levels of craving have been associated with the increased probability of alcohol relapse (Bottlender and Soyka 2004; Schneekloth et al. 2012). Current pharmacological options, including acamprosate (Witkiewitz et al. 2012), naltrexone (Rösner et al., 2010) and nalmefene (Paille and Martini 2014), have targeted reducing cravings with the aim of reducing relapse. Another approach includes mindfulness interventions which may provide a further approach to support treatment through increased acceptance and acting with awareness (Bowen et al. 2009).

A number of psychosocial factors, including quality of life (QOL) and sleep, are disrupted by AUD and may be further indicators of the capacity of MDMA-assisted psychotherapy to improve outcomes in AUD. Numerous studies have found that AUD patients report impaired QOL compared with the general population, particularly in emotional and mental health and social functioning (Ugochukwu et al. 2013). Moreover, studies have also revealed that AUD is both a risk factor and a primary cause of low QOL (Ugochukwu et al. 2013). Higher alcohol consumption or longer heavy drinking days is directly correlated with lower levels of QOL. Importantly, studies have reported that alcohol abstinence and its maintenance improve QOL across multiple domains in patients (Volk et al. 1997, Witkiewitz et al. 2018). The relationship between QOL deterioration and AUD has been found to be moderated by many factors, including comorbidity with psychiatric disorders in particular depression (Witkiewitz et al. 2018). Similarly, sleep disturbances are both common and impactful in AUD. Insufficient sleep is associated with lower levels of health and QOL (Strine and Chapman 2005) and disturbed sleep and insomnia are common in AUD (Koob and Colrain 2020). Insomnia is highly prevalent in long-term abstinent AUD individuals and has been linked to an increase in the likelihood to relapse (Brower 2015).

Self-compassion and empathy are two key psychological processes that have been shown to be impacted by MDMA (Kamboj et al. 2015, Carlyle et al. 2019). Self-compassion contributes mood, health, and well-being (Zessin et al. 2015) and changes in self-compassion and empathy may be transdiagnostic mediators of treatment effects in addiction and other mental health disorders. AUD impacts several facets of self-compassion with evidence demonstrating significantly lower levels of mindfulness, common humanity and self-kindness, and significantly higher levels in over-identification, perceived isolation and self-judgement than the norms for general population (Brooks et al. 2012). Social impairments are frequently observed in individuals with AUD, and empathy is a crucial component for forming and maintaining social relationships (Kumar et al. 2022). Therapeutic approaches aimed at enhancing self-compassion and empathy could potentially lead to improved outcomes for individuals with AUD. Additionally, psychotherapeutic methods that emphasize compassion and mindfulness may contribute to enhancing self-compassion and reducing addictive behavioral processes which may reduce substance misuse (Luoma and Platt 2015, Garland and Howard 2018).

The safety and tolerability from the first trial of MDMA-assisted psychotherapy in AUD has been previously published (Sessa et al. 2021). Here, we perform a Bayesian analysis to quantify treatment success, defined as a two-level reduction in WHO drinking risk at the 3-month follow-up. Furthermore, we present the secondary outcome measures of drinking behavior, QOL, sleep, self-compassion, and empathy.

Materials and methods

Approvals and drug source

This trial was sponsored by Imperial College London. The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Helsinki Declaration of 1975, as revised in 2008. All procedures involving human subjects/patients were approved by the South West—Central Bristol Research Ethics Committee of the National Research Ethics Service (IRAS ID: 198906, REC Reference: 16/SW/0273) and the Medicines and Healthcare products Regulatory Agency (MHRA). Good Manufacturing Practice MDMA was obtained from Sterling Pharmaceuticals (Newcastle) and formulated into the investigational medicinal product (62.5 mg MDMA in gelatin capsules) by the Pharmacy Manufacturing Unit at Guy's and St Thomas' NHS Foundation Trust (London, UK). Awakn Life Sciences Corporation have exclusive rights to the data from the Phase 2b. ClinicalTrials.gov ID: NCT04158778.

Participants

Patients with AUD took part in a safety and tolerability study of MDMA-assisted psychotherapy. Written informed consent was obtained from all participants. Full study details and findings on the primary outcomes are reported in the original publication (Sessa et al. 2021).

Procedure

Full details of the study procedures have been reported elsewhere (Sessa et al. 2021). In summary, this was an open-label, safety and tolerability feasibility study in 14 patients aged 18 study details and finding recently undergone detoxification. All patients received MDMA-assisted psychotherapy. Primary outcome measures were the number of patients completing the eight-week psychotherapy course, the number accepting the second booster dose of MDMA on drug-assisted days and adverse events, all of which have been previously reported

(Sessa et al. 2021). Secondary outcome measures included changes in drinking behavior, measures of mental well-being, QOL, sleep, and psychosocial functioning.

Participants received an eight-week recovery-based therapy course, which included ten psychotherapy sessions. Two of these sessions (sessions 3 and 7) involved dosing with MDMA, with an initial oral dose of 125 mg followed by a 62.5 mg booster dose, while other sessions comprised of psychotherapy without MDMA.

WHO risk drinking level

Units of alcohol consumed was measured by the Timeline Followback (TLFB). We analysed the number of units of alcohol consumed to determine whether a 2-level reduction in WHO risk drinking level occurred. We performed Bayesian analysis upon this classification.

PROMs

Several Patient Reported Outcome Measures (PROMs) were collected using standardized and well validated questionnaires stated below.

Drinking behavior measures

Penn alcohol craving scale

The (Penn alcohol craving scale) PACS is a five-item, self-report measure that includes questions about the frequency, intensity, and duration of craving, the ability to resist drinking, and asks for an overall rating of craving for alcohol for the previous week (Flannery et al. 1999). Each question is scaled from 0 to 6. The total PACS score is presented (lower scores indicate less craving).

Obsessive compulsive drinking scale

The obsessive compulsive drinking scale (OCDS) is a 14-item questionnaire that measures an individual's alcohol use and their attempts to control their drinking (Anton et al. 1995, 1996). Total scores for the obsessive subscale and compulsive subscale are presented (lower scores indicate less severe obsessive and compulsive drinking behavior).

Quality of life, sleep, and psychosocial functioning measures

Quality of life

The 36 item short form (SF-36) was used as a measure of QOL (Ware and Sherbourne 1992, McHorney et al. 1994). It comprises 36 questions that cover eight domains of health; limitations in physical activities because of health problems, limitations in social activities because of physical or emotional problems, limitations in usual role activities because of physical health problems, bodily pain, general mental health (psychological distress and well-being), limitations in usual role activities because of emotional problems, vitality (energy and fatigue), general health perceptions. Each scale is directly transformed into a 0–100 scale on the assumption that each question carries equal weight. The lower the score the more impairment in that domain.

Sleep

The quality of sleep was assessed using four items derived from the Pittsburgh Sleep Quality Index (PSQI) (Buysse et al. 1989), each of which was assessed on a five-point scale: (i) how much sleep patterns were interfering with daily activities (not at all, a little, somewhat, much, very much); (ii) difficulties falling asleep, (iii) difficulties staying asleep and; (iv) waking up too early were each assessed using the five-point scale: none, mild, moderate, severe, very severe. In the current analyses the four 5-item variables are combined to form a continuous measure of sleep disturbance. The total PSQI score is presented (a lower score indicates less impaired sleep).

Self-compassion scale

The scale consists of 26 items to measure the three main components of self-compassion on separate subscales (self-kindness versus self-judgement, common humanity versus isolation, and mindfulness versus over-identification) (Neff 2003). The total self-compassion scale (SCS) score is presented (a lower score indicates lower levels of self-compassion).

Interpersonal reactivity index (IRI)

The interpersonal reactivity index (IRI) scale consists of 28-item comprised of four 7-item subscales: perspective taking (tendency to adopt the psychological point of view of others); fantasy (tendency to transpose oneself imaginatively into the feelings and actions of fictitious characters in books, movies, and plays), empathic concern (feelings of sympathy and concern for others) and personal distress (measures 'self-oriented' feelings of personal anxiety and unease in tense interpersonal settings) (Davis 1980). A lower score in each subscale indicates lower levels for that tendency/measure.

Data analysis

For the binary outcome measure, i.e. a 2-level reduction in WHO drinking risk levels, we used binomial test, which we followed up by a Bayesian analysis. Bayesian analysis is an alternative statistical method to the commonly used frequentist approaches, often providing a more clinically informative analysis of treatment impacts (Nayak et al. 2023). This view is supported by a systematic review of Bayesian reanalyses involving 82 studies from high-impact critical care journals revealed that instances of disagreement between frequentist and Bayesian outcomes are infrequent (Yarnell et al. 2021). Furthermore, there are exemplars within cannabis and psychedelic research that reveal benefits of employing Bayesian analysis (Nayak et al. 2023, Szigeti et al. 2023).

We used two priors for the Bayesian analysis, both from the beta distribution family. We used a flat prior ($a = 1$, $b = 1$; i.e. mean success rate is 50%) and skeptical ($a = 1$, $b = 3$; i.e. mean success rate is 25%) priors. This choice of the skeptical prior was driven by that in mental health treatments the response rates in placebo conditions are typically in the 25–40% range (Bschor et al. 2024). We took the lower end of this range to define the mean of the skeptical prior. In other words, if it is assumed that a treatment has no specific effect, which represent a skeptical attitude, even then at least 25% success rate is expected.

All PROMs data were continuous outcomes and these were analysed with Jamovi (Version 2.3) and graphs were created using GraphPad Prism 9 (GraphPad Software Inc.). Data on graphs are presented as mean ± standard error of mean (SEM). Due to the small population, Wilcoxon matched-pairs signed rank tests were performed for non-parametric data. We provide 95% confidence intervals (CI) around the mean differences. Bonferroni corrected p values were used for PROMs with multiple subscales. We calculated effect sizes and present the rank biserial correlation in the appendix.

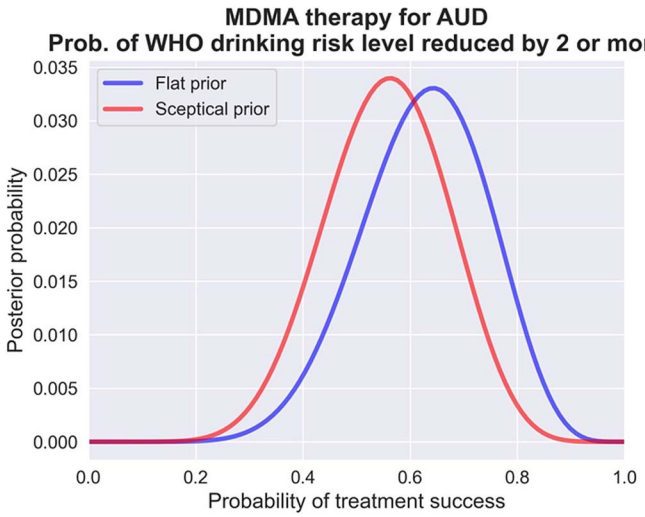


Figure 1 Posterior probability of treatment success of MDMA therapy for AUD at 3 months follow-up. Results are shown for priors that are flat (any success rate between and including 0% and 100% is possible) and skeptical (gentle, excluding 0% and 100%, with a mean at 25%) ($n = 14$).

Table 1. Mean posterior probability of treatment success of MDMA therapy for AUD at 3 months follow-up

Outcome	Sample size	Success rate	Flat prior		Skeptical prior	
			Credible interval	Mean posterior	Credible interval	Mean posterior
WHO risk drinking level reduced by 2	14	9/14	0.4–0.85	0.63	0.33–0.77	0.55

Results

Demographics

The study enrolled 14 participants (8 males and 6 females; mean age = 48 years). All were white British. Full demographics are reported in the original publication (Sessa et al. 2021).

Bayesian analysis of World Health Organization drinking risk level at 3 months follow-up

We defined treatment success as a 2-level reduction in WHO drinking risk to perform Bayesian analysis (Fig. 1, Table 1). Using this definition of success, binomial test did not reject the null hypothesis when testing against an expected success rate of 50% (k = .64, $P = .423$); the 95% confidence interval of success rate was 0.35–0.87.

We followed up this initial test with Bayesian analysis for better interpretability. Assuming a flat prior, meaning that, prior to the treatment there is equal belief that the treatment will be success/failure, the probability of success of treatment is 63% with credible intervals of 40% to 85%. Assuming a skeptical prior, a gentle distribution excluding 0% and 100%, with a mean at 25%, the probability of treatment success is 55% with credible intervals of 33% to 77%, see Fig. 1, Table 1.

Drinking behavior at 3 months follow-up

Drinking behavior, including assessments of alcohol craving, was measured using the PACS and OCDS (Fig. 2, full statistical details in the Supplementary Materials Table S1–S4). Significant reductions in the total PACS score were observed at 3 months follow-up compared to baseline (difference 5.50 [95% CI .5–13], $P = .031$, $r_b = .781$). There were significant reductions in the obsessive (difference 5.00 [95% CI 2.00–8.50], $P = .006$, $r_b = .868$) and compulsive subscales (difference 5.00 [95% CI 1.00–9.00], $P = .021$, $r_b = .769$) of the OCDS at 3 months compared to baseline.

Quality of life, sleep, well-being, and psychosocial functioning measures

QOL, sleep, and psychosocial functioning were assessed using PROMs (Fig. 3, full statistical details in the Supplementary Materials Table S5–S12).

Quality of life

There was a general trend of an increase in several aspects of QOL (Fig. 3, Supplementary Materials Table S5–S6). Although there were significant increases for several subscales (including energy/fatigue, emotional well-being, social functioning, pain, general health, and change in reported health) at 3 months compared to baseline, once the p values were Bonferroni corrected no subscales retained significance.

Sleep

There was a significant reduction in total PSQI score indicating improvements in sleep at 3 months compared to baseline (difference 2.00 [95% CI 7.54e-6-3.00], $P = .041$, $r_b = .648$) (Fig. 3, Supplementary Materials Table S7–S8).

Self-compassion

A significant increase in the total self-compassion score was observed at 3 months follow-up compared to baseline (difference 12.1 [95% CI 9.09–14.4], $P < .001$, $r_b = 1.00$) indicating improvements in self-compassion (Fig. 3, Supplementary Materials Table S9–S10).

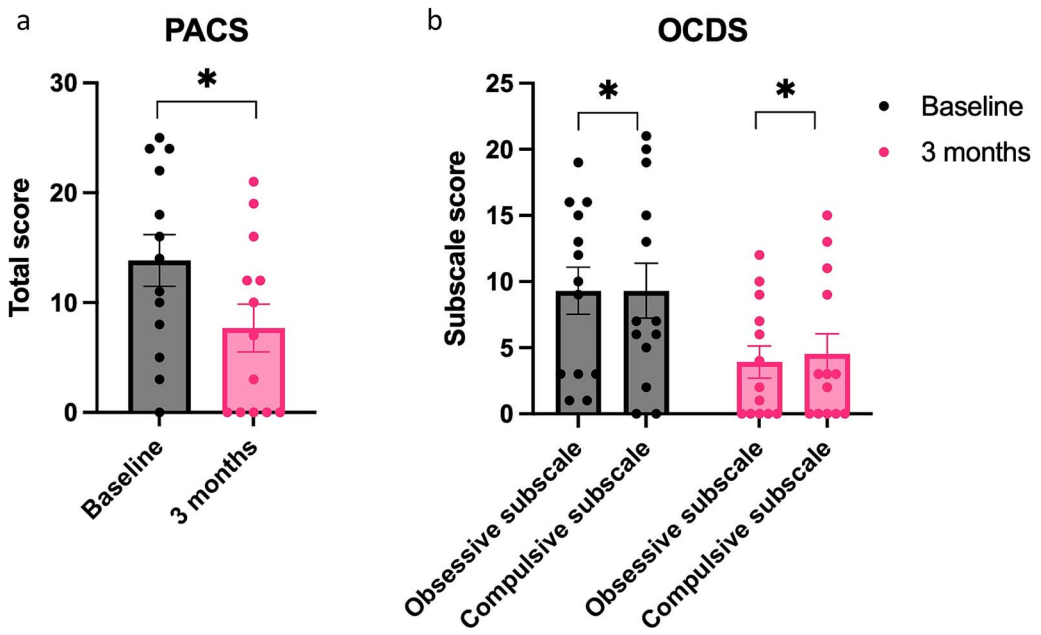


Figure 2 PACS (a) and OCDS (b) scores at baseline and 3 months follow-up after MDMA-assisted psychotherapy for AUD. Data are represented as mean ± SEM ($n = 13$). Data were analysed using Wilcoxon matched-pairs signed rank test (Bonferroni corrected P-values were used for PROMs with multiple subscales). PACS: Alcohol craving scale; OCDS: Obsessive Compulsive Drinking Scale. $^*P < .05$

Empathy

There was only significant reductions in the personal distress subscale at 3 months compared to baseline (difference .5 [95% CI .143–0.929], $P = .009$, $r_b = .945$), indicating improvements in levels of personal distress (Fig. 3, Supplementary Materials Table S11–S12). For the IRI subscales, mean differences were calculated as baseline score minus 3-month score, such that positive values indicate a reduction in score over time.

Discussion

We present a Bayesian reanalysis of data from a previously published feasibility study of MDMA-assisted psychotherapy in AUD. Treatment success was defined as 2-level reduction in WHO drinking risk. Our analysis estimates that there is a probability of treatment success of 55%–63%, dependent upon a flat or skeptical prior respectively. Additionally, we present preliminary findings indicating reductions in alcohol craving, improvements in sleep, and self-compassion, as well as a trend toward significant improvements in several aspects of QOL.

The European Medicines Agency (EMA) recommends abstinence as a primary endpoint, but also accepts intermediate harm reduction endpoints, including reductions in total alcohol consumption, reduction in heavy drinking days, or reduction in WHO drinking risk levels (European Medicines Agency 2010). The inclusion of harm reduction endpoints other than abstinence is important because most individuals who receive treatment for AUD have difficulty sustaining abstinence and many prefer drinking reduction goals. Reductions in WHO risk levels for alcohol consumption appear to be achievable, associated with better functioning, and maintained over time in both the United States and the United Kingdom (Witkiewitz et al. 2020). Beyond meaningful improvements in functioning, physical health, and QOL, a reduction of at least two levels in WHO risk drinking categories is also linked to lower healthcare costs over 1 and 3 years (Aldridge et al. 2022). The WHO Risk Drinking Levels of Alcohol Consumption has been accepted within the process of the FDA COA Qualification Program, but it yet to be accepted as a Full Qualification Package (FQP) (FDA 2020). Our analysis estimates treatment success of 55–63% for a 2-level reduction in WHO drinking risk at 3 months post-MDMA assisted therapy, which is consistent with success reported in other AUD trials (Witkiewitz et al. 2021). Specifically, secondary data analysis of patients with AUD enrolled in the two large multi-center trials, COMBINE Study and Project MATCH, report two-level WHO drinking risk reductions achieved by 68% of patients at the end of treatment, with 77.7% of those individuals maintaining that reduction at a 3-year follow-up (Witkiewitz et al. 2021). Although the findings reported in this paper are preliminary and of a small sample size, the treatment success of MDMA-assisted psychotherapy (55%–63%) is comparable with those reported for current behavioral interventions and/or non-psychedelic drugs for AUD as demonstrated by the secondary analysis from the COMIBINE study and Project MATCH (68%). Importantly, these results suggest that MDMA-assisted psychotherapy may produce clinically meaningful reductions in alcohol consumption in a manner comparable with established treatments, warranting further investigation in larger trials.

Understanding the relationship between craving and alcohol use after treatments can be useful for treatment planning and modulating intervention approaches (Anton 2013). The OCDS is considered a robust predictor of alcohol use in the following week after assessment (Roberts et al. 1999) but not in longer term periods (Kranzler et al. 1999). Secondary analysis from the COMBINE trial demonstrated that craving for alcohol, assessed by a 3-item self-report Craving Scale, is strongly associated with alcohol use in the following week (McHugh et al. 2016). We show a reduction in craving assessed using the PACS and OCDS. Future studies

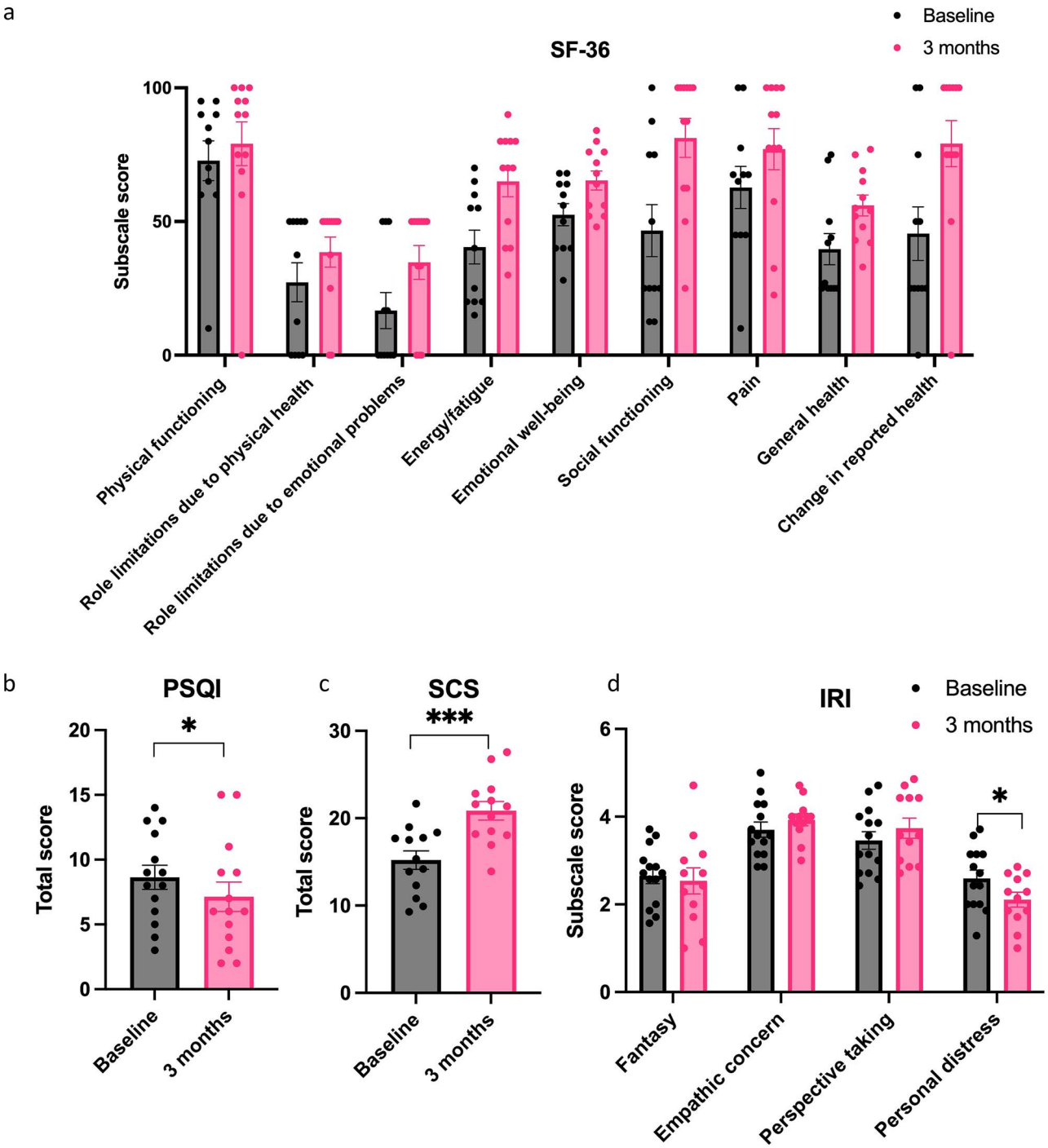


Figure 3 PROMs across QOL (a), sleep (b), self-compassion (c) and interpersonal reactivity (d) at baseline and 3 months follow-up after MDMA-assisted psychotherapy for AUD. Data are represented as mean ± SEM. Data were analysed using Wilcoxon matched-pairs signed rank test (Bonferroni corrected *P*-values were used for PROMs with multiple subscales). SF-36: 36 item short form ($n = 11$); PSQI: Pittsburgh Sleep Quality Index ($n = 14$); SCS: self-compassion scale (n = 13); IRI: Interpersonal reactivity index ($n = 12$). $^{*}P < .05$, $^{**}P < .01$, $^{***}P < .001$.

should determine whether craving mediates or predicts outcome response in the context of MDMA-assisted psychotherapy.

There is growing recognition of the value and importance of QOL, as well as broader well-being, in healthcare and research (Haraldstad et al. 2019), including in the context of AUD (Donovan et al. 2005). QOL and well-being measures provide insights into the impact of a treatment beyond specific symptom relief, reflecting daily functioning and the perceived ability to lead a fulfilling life (Haraldstad et al. 2019). Furthermore, sleep contributes to QOL and general well-being (Strine and Chapman 2005). Therefore, the assessment of QOL and sleep in AUD studies provides a more comprehensive understanding on the impact of treatment beyond just drinking behavior. We show a trend toward significant improvements in several aspects of health, specifically energy/-fatigue, emotional well-being, social functioning, pain, general health, change in reported health. The improvements in social

functioning are of relevance as AUD can lead to substantial problems in interpersonal relationships (Levola et al. 2014).

Acutely MDMA produces subjective prosocial feelings (Dumont et al. 2009), as well as enhances empathy and sociability, which are thought to contribute to its role in psychotherapy (Bedi et al. 2010, Hysek et al. 2012, Mithoefer et al. 2013). These acute effects appear to persist as we found improvements in self-compassion and personal distress on the IRI scale. This is in-line with improvements in emotional empathy and self-compassion have been reported in recreational MDMA use (Kamboj et al. 2018). Further studies should establish the role of self-compassion and empathy in MDMA-assisted psychotherapy for AUD.

Limitations

This trial had a number of limitations, some of which have already been discussed (Sessa et al. 2021). This was open-label trial with a small population and therefore there is a need for larger and randomized placebo-controlled trials to assess efficacy. Given the exploratory nature of this study and the number of secondary outcomes assessed, we acknowledge the potential for Type I error inflation. As such, these findings should be interpreted with caution and considered hypothesis-generating rather than definitive evidence of efficacy. Future studies with larger samples and appropriate corrections for multiple comparisons will be necessary to confirm these preliminary results.

The blinding of MDMA is challenging and remains a challenge in other psychedelic research (Muthukumaraswamy et al. 2021). Therefore, further research should measure deblinding and expectancy. Additionally, there is also the need for long-term follow-up data to assess durability of treatment in AUD. This study's enrolment criteria excluded participants using antidepressants (unless titrated off prior to commencement of the trial), serious suicide risk, comorbid psychotic/personality disorders and underlying cardiovascular and severe liver disease. Future studies should carefully consider exclusion criteria as these restrictions may limit generalizability of results, especially the exclusion of those using antidepressants and not willing to cease use, considering the co-morbidity between AUD and depression. This study lacked ethnic diversity and future research must support recruitment of ethnic diverse populations, as such in the MAPS Phase 3 confirmatory trial (Mitchell et al. 2023). A further limitation of this study is that SF-36 measures broad aspects of QOL that may be influenced by factors beyond alcohol use, including comorbid conditions. However, due to the small sample size and lack of systematically collected comorbidity data, we were unable to account for these potential confounds. Future studies with larger samples and detailed comorbidity assessments would be better suited to investigate this.

Future directions

Participants received an eight-week course of relapse-prevention therapy comprising 10 psychotherapy sessions, employing aspects of motivational interviewing and 'third-wave' cognitive–behavioral approaches. Manualized approaches to psychotherapy have been employed in MDMA-assisted psychotherapy for PTSD (Mitchell et al. 2021, 2023) and ketamine-assisted psychotherapy for AUD (Grabski et al. 2022). Further larger scale trials of MDMA-assisted therapy in AUD will need to determine the therapeutic approach and whether manualized approaches should be used. Training of therapists to provide MDMA-assisted psychotherapy will present a hurdle for uptake into healthcare and presents a limiting factor for wide-spread availability (Madero and Alvarez 2023), so manualized approaches may assist in roll out.

Another hurdle for MDMA-assisted psychotherapy entering into healthcare systems is the 8-hour dosing treatment session and the requirement of two therapists to be present in these (Halvorsen et al. 2021). Also, for safety reasons in the current trial, participants remained overnight in the treatment center after each drug-assisted session, though this might not be needed in future given few adverse effects emerged during the night. Nevertheless, shorter forms of MDMA therapy, similar to current models of ketamine-assisted psychotherapy, would be more cost-effective. This could be provided either by a different formulations or delivery routes of MDMA.

Conclusion

The safety and tolerability of MDMA-assisted psychotherapy in AUD had been previously reported. This paper builds on that foundation by analysing secondary outcomes, providing additional insights relevant to regulatory discussions in an evolving landscape of MDMA-assisted psychotherapy and future clinical implementation. The Bayesian analyses presented here provides a useful alternative perspective on reductions in drinking in terms of a 2-level reduction in WHO drinking risk, indicating estimates for treatment success at 55%–63%. The findings also provide preliminary insights into the potential impact of MDMA-assisted psychotherapy on drinking craving, sleep, QOL, and general well-being beyond reductions in drinking, which may support long-term recovery in individuals with AUD. These results provide a good foundation for measures that could be utilized in future placebo-controlled clinical trials to determine the efficacy of MDMA-assisted psychotherapy in treating AUD.

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Author contributions

Hannah Thurgur (Conceptualization [lead], Formal analysis [lead], Writing—original draft [lead], Writing—review & editing [lead]), Ben Sessa (Conceptualization [supporting], Writing—review & editing [supporting], Investigation [supporting]), Laurie Higbed (Conceptualization [supporting], Writing—review & editing [supporting], Investigation [supporting]), Steve O'Brien (Writing—review & editing [supporting], Investigation [supporting]), Balázs Szigeti (Formal analysis [supporting], Writing—review & editing [supporting]), Celia J. A. Morgan (Writing—review & editing [supporting]), David J. Nutt (Conceptualization [supporting], Funding acquisition [lead], Writing—review & editing [supporting]), Claire Durant (Investigation [supporting]), and Sue Wilson (Investigation [supporting]). All authors read and approved the submitted version.

Supplementary data

Supplementary data is available at *Alcohol and Alcoholism* online.

Conflict of interest: HT was a Senior Research Officer at Drug Science during the production of the manuscript where as part of her position she collaborated with industry partners who are researching the development of medical psychedelics and was seconded to Awakn Life Sciences. HT is no longer in these positions at the point of submission. DJN is CSO of Awakn Life Sciences. CJAM was seconded to Awakn Life Sciences and University of Exeter has licensed IP generated from her work to Awakn at the time of manuscript production. BS, LH, SO, CD, SW, BS declare no conflicts of interest.

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Data availability

The anonymized data are available on reasonable request from the corresponding author (HT).

Ethics statement

This trial was sponsored by Imperial College London, approved by the Central Bristol Research Ethics Committee of the National Research Ethics Service and the Medicines and Healthcare products Regulatory Agency (MHRA).

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